

Does an AGN's infrared spectrum remember its accretion history?

Scientific Category: Supermassive Black Holes and Active Galaxies

Alternate Category: Interstellar Gas and Dust

Scientific Keywords: AGN host galaxies, Interstellar dust, Quasars

Instruments: MIRI

Proposal Size: Medium

Exclusive Access Period: 12 months

Allocation Information (in hours):

Science Time: 17.1

Charged Time: 67.0

Abstract

An AGN can fade while its warm dust still glows from an earlier bright state, and after it brightens the dust can lag behind the new nuclear power. How long does this infrared memory last, and is dust replenished after a nucleus fades? We request 67.0 hours of MIRI/MRS spectroscopy of 24 AGN at $z < 0.3$ with recorded rises, declines, flares and recoveries. ZTF and decade-long WISE monitoring, SPHEREx measurements and multi-epoch optical spectroscopy tie each nuclear history to its hot dust. MIRI supplies what no other facility now can: the 5-28 micron warm continuum, both silicate features, and the PAH and spatial diagnostics that separate the nucleus from its host. Anchored by 2010 WISE 12 and 22 micron photometry, we will jointly fit each spectrum with its activity history and test whether nuclei at comparable present power carry different warm spectra because they reached that power by different paths. Delayed heating and cooling of a fixed dust distribution is the null hypothesis. Fading and recovering sources test whether dust must be destroyed, replenished or redistributed; rising sources provide the comparison. Together the 24 spectra constrain the response timescales and the radial extent of the responding dust. The outcome is a bound on how fast the dusty structure around an AGN can change, and a measure of how long the mid-infrared emission of a changed nucleus misrepresents its present state, which matters for surveys that use mid-infrared light to gauge AGN power and obscuration.

Target Summary:

Target	RA	Dec
F01	07 59 34.9500	+32 21 43.30
F01-SKY	07 59 39.6840	+32 19 59.37
F02	00 33 52.7100	+20 21 4.10
F02-SKY	00 33 56.8310	+20 21 19.63
F03	10 53 25.4100	+30 24 19.30
F03-SKY	10 53 33.4960	+30 24 28.44
F04	00 53 46.1600	+22 32 22.40
F04-SKY	00 53 43.9190	+22 30 26.49
F05	08 53 59.0600	+21 28 3.40
F05-SKY	08 54 1.2850	+21 29 59.31
F06	01 25 58.0600	+15 18 13.70
F06-SKY	01 25 55.1270	+15 18 56.13
F07	23 44 33.0900	+26 00 38.90
F07-SKY	23 44 38.1330	+26 01 10.59
F08	10 44 27.7400	+27 18 6.50
F08-SKY	10 44 32.0880	+27 18 22.02
F09	08 35 38.7800	-04 05 17.50
F09-SKY	08 35 34.7700	-04 05 17.50
F10	08 53 37.2600	+01 43 3.60
F10-SKY	08 53 30.9130	+01 42 19.22
F11	23 30 32.3000	-02 27 45.00
F11-SKY	23 30 29.2970	-02 26 27.06
F12	04 58 40.3000	-21 59 31.00
F12-SKY	04 58 35.4470	-22 00 51.43
R01	00 46 7.9766	+09 07 20.95
R01-SKY	00 46 5.2050	+09 09 13.71
R02	22 05 37.7100	-07 11 14.50
R02-SKY	22 05 29.9220	-07 10 43.44
R03	10 47 5.1600	+54 44 5.90
R03-SKY	10 46 59.9610	+54 45 23.84
R04	08 29 42.6700	+41 54 36.90
R04-SKY	08 29 33.3630	+41 53 36.88
R05	10 40 43.3900	+27 35 19.00
R05-SKY	10 40 40.1990	+27 34 36.57
R06	08 38 26.5000	+37 19 6.70
R06-SKY	08 38 22.6460	+37 19 45.26

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Target	RA	Dec
R07	10 11 52.9800	+54 42 6.40
R07-SKY	10 11 52.9800	+54 43 6.40
R08	09 48 38.4200	+40 30 43.60
R08-SKY	09 48 42.4500	+40 30 5.03
R09	06 55 49.5300	+40 00 1.20
R09-SKY	06 55 42.7480	+39 59 16.19
R10	19 26 30.2200	+41 33 5.00
R10-SKY	19 26 34.0530	+41 34 6.43
R11	19 49 13.7000	-18 47 16.00
R11-SKY	19 49 17.9090	-18 47 21.23
R12	20 14 59.2600	+25 23 1.20
R12-SKY	20 15 0.1280	+25 20 46.71

Observing Summary:

Target	Observing Template	Flags	Allocation *
F01	MIRI Medium Resolution Spectroscopy		1,268 / 6,492
F01-SKY	MIRI Medium Resolution Spectroscopy		1,268 / 3,883
F02	MIRI Medium Resolution Spectroscopy		1,268 / 5,064
F02-SKY	MIRI Medium Resolution Spectroscopy		1,268 / 3,819
F03	MIRI Medium Resolution Spectroscopy		1,268 / 4,969
F03-SKY	MIRI Medium Resolution Spectroscopy		1,268 / 3,870
F04	MIRI Medium Resolution Spectroscopy		1,268 / 6,512
F04-SKY	MIRI Medium Resolution Spectroscopy		1,268 / 3,883
F05	MIRI Medium Resolution Spectroscopy		1,268 / 6,609
F05-SKY	MIRI Medium Resolution Spectroscopy		1,268 / 3,883
F06	MIRI Medium Resolution Spectroscopy		1,268 / 6,609
F06-SKY	MIRI Medium Resolution Spectroscopy		1,268 / 3,819
F07	MIRI Medium Resolution Spectroscopy		1,268 / 6,609

Does an AGN's infrared spectrum remember its accretion history?

Target	Observing Template	Flags	Allocation *
F07-SKY	MIRI Medium Resolution Spectroscopy		1,268 / 3,838
F08	MIRI Medium Resolution Spectroscopy		1,268 / 4,843
F08-SKY	MIRI Medium Resolution Spectroscopy		1,268 / 3,819
F09	MIRI Medium Resolution Spectroscopy		1,268 / 6,512
F09-SKY	MIRI Medium Resolution Spectroscopy		1,268 / 3,819
F10	MIRI Medium Resolution Spectroscopy		1,268 / 5,229
F10-SKY	MIRI Medium Resolution Spectroscopy		1,268 / 3,870
F11	MIRI Medium Resolution Spectroscopy		1,268 / 6,512
F11-SKY	MIRI Medium Resolution Spectroscopy		1,268 / 3,855
F12	MIRI Medium Resolution Spectroscopy		1,268 / 6,609
F12-SKY	MIRI Medium Resolution Spectroscopy		1,268 / 3,870
R01	MIRI Medium Resolution Spectroscopy		1,268 / 6,512
R01-SKY	MIRI Medium Resolution Spectroscopy		1,268 / 3,883
R02	MIRI Medium Resolution Spectroscopy		1,268 / 6,609
R02-SKY	MIRI Medium Resolution Spectroscopy		1,268 / 3,883
R03	MIRI Medium Resolution Spectroscopy		1,268 / 6,492
R03-SKY	MIRI Medium Resolution Spectroscopy		1,268 / 3,855
R04	MIRI Medium Resolution Spectroscopy		1,268 / 4,949
R04-SKY	MIRI Medium Resolution Spectroscopy		1,268 / 3,883
R05	MIRI Medium Resolution Spectroscopy		1,268 / 6,609
R05-SKY	MIRI Medium Resolution Spectroscopy		1,268 / 3,819
R06	MIRI Medium Resolution Spectroscopy		1,492 / 6,752
R06-SKY	MIRI Medium Resolution Spectroscopy		1,492 / 4,079
R07	MIRI Medium Resolution Spectroscopy		1,268 / 5,062

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Target	Observing Template	Flags	Allocation *
R07-SKY	MIRI Medium Resolution Spectroscopy		1,268 / 3,819
R08	MIRI Medium Resolution Spectroscopy		1,268 / 6,512
R08-SKY	MIRI Medium Resolution Spectroscopy		1,268 / 3,819
R09	MIRI Medium Resolution Spectroscopy		1,268 / 6,512
R09-SKY	MIRI Medium Resolution Spectroscopy		1,268 / 3,855
R10	MIRI Medium Resolution Spectroscopy		1,268 / 6,609
R10-SKY	MIRI Medium Resolution Spectroscopy		1,268 / 3,838
R11	MIRI Medium Resolution Spectroscopy		1,268 / 6,609
R11-SKY	MIRI Medium Resolution Spectroscopy		1,268 / 3,819
R12	MIRI Medium Resolution Spectroscopy		1,268 / 6,512
R12-SKY	MIRI Medium Resolution Spectroscopy		1,268 / 3,896

* Science duration / charged duration (sec)

Total Prime Science Time in Hours: 17.1

Total Charged Time in Hours: 67.0

Observation Description, Investigators, and Team Expertise are all included in this preview for your team to review. These will not appear in the version of the proposal given to the TAC. The Observation Description is only needed by STScI operations, and the investigator information is removed to allow for a dual anonymous review.

Observing Description

We request 67.0 hours of MIRI Medium Resolution Spectroscopy (MRS) for 24 nearby ($z < 0.3$) changing-look and strongly variable AGN with recorded rises, declines, flares and recoveries. Each nucleus will be observed once in all three MRS grating settings, providing approximately 4.9–27.9 micron coverage of the warm continuum, both silicate features, PAH bands and diagnostic emission lines. Science observations use nuclear target acquisition, four-point dithers and FULL/FASTR1 readout. The requested continuum sensitivity is $S/N = 50$ per $R = 100$ bin near rest-frame 5 and 12 microns and $S/N = 30$ near 18 microns. The faintest W3 target, J083826.50+371906.7, receives additional exposure in setting B.

Each science visit has an equal-depth observation of a nearby blank-sky field selected using WISE and 2MASS data. Separate sky observations preserve extended host emission for spatial and spectral decomposition. Sky visits use four-point extended-source dithers without target acquisition. Each science/sky pair is linked as a consecutive, noninterruptible sequence to provide a contemporaneous background. The programme comprises 24 science and 24 sky visits, with 17.00 hours of science-plus-sky integration and 67.0 hours total charged time. No fixed-date constraints, coordinated parallels or joint-facility time are requested.

Investigators:

Role	Investigator	Institution	Country
CoI	Dr. Andreas L Faisst	California Institute of Technology	USA/CA
CoI	Dr. George Helou	California Institute of Technology	USA/CA
PI &	Dr. Shoubaneh Hemmati	California Institute of Technology	USA/CA
CoI	Dr. Jessica Krick	California Institute of Technology	USA/CA
CoI	Dr. Yu-Heng Lin	California Institute of Technology	USA/CA
CoI *	Dr. Irene Shivaiei	Centro de Astrobiologia - CAB	ESP
CoI	Dr. David L Shupe	Caltech/IPAC	USA/CA
CoI	Dr. Wuji Wang	California Institute of Technology	USA/CA

Number of investigators: 8

* ESA investigators: 1

& Contacts: 1

Team Expertise:

Our team combines AGN variability, infrared dust physics, MIRI/MRS spectroscopy and the calibration of the surveys that supply the activity histories. The ancillary data are in hand: ZTF and WISE/NEOWISE light curves for all 24 targets, SPHEREx spectra already extracted for the sample, archival multi-epoch optical spectra, and Palomar/NGPS spectra obtained in 2026. Most of the team is based at Caltech/IPAC, which hosts the IRSA archives for WISE, ZTF and SPHEREx.

PI Shoubaneh Hemmati developed the machine-learning light-curve analysis used to select the sample (Hemmati et al. 2026, ApJ, 998, 130), leads the Palomar/NGPS follow-up, and will coordinate reconstruction of

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the activity histories and the joint tests of dust memory and evolution.

Irene Shivaiei served on the JWST/MIRI development and commissioning teams and leads MIRI/MRS analysis of PAH emission in the JWST program PAHSPECS; she will guide MRS reduction, spectral decomposition and interpretation of resolved host dust.

George Helou brings decades of work on infrared emission, dust heating and the separation of star-formation and AGN contributions, and will lead the physical interpretation of the warm continuum and aromatic features together with the dust-response modelling.

Jessica Krick brings Spitzer/IRAC calibration expertise, including infrared photometry of JWST calibration stars, and will lead the synthetic-photometry comparison with the 2010 WISE baseline and the assessment of background and calibration uncertainties.

Andreas Faisst, a SPHEREx science-team member with dust-modelling experience and a leading role in COSMOS-Web, brings expertise in MIRI/MRS data analysis and leads JWST/MIRI programs on PAH emission. He will validate the SPHEREx spectra and support joint modelling of nuclear and host emission.

David Shupe brings ZTF pipeline experience in astrometric calibration, source matching and relative photometry, and will handle light-curve quality control and reproducible analysis.

Yu-Heng Lin brings quasar spectroscopy and emission-line diagnostics, and will measure the optical spectral states for comparison with the infrared diagnostics.

Wuji Wang brings expertise in AGN studies and integral-field spectroscopy, both central to this proposal. He has led several JWST IFU programs, providing directly relevant experience for the MIRI/MRS observations proposed here.

■ Scientific Justification (required for all)

Does the dust around a changing nucleus keep up with it? An active galactic nucleus (AGN) can change luminosity by an order of magnitude within years [1,2], sometimes transforming its optical continuum and broad emission lines [3–5]. Its dust records these changes through delayed heating, sublimation, or replenishment. Individual nuclei show dust boundaries adjusting over years [6,7]; what is missing is an ensemble test of how the response depends on the direction, amplitude, and timing of the transition.

We request **67.0 h of MIRI/MRS spectroscopy of 24 AGN at $z < 0.3$** caught at distinct transition stages. **The experiment pairs one warm-dust spectrum per nucleus with its recorded activity history to test whether delayed heating alone explains the dust we see.** ZTF/NEOWISE light curves, SPHEREx coverage, and optical spectroscopy through 2026 trace the central engine and hot inner boundary; MRS supplies the warm continuum and both silicate profiles needed to distinguish a lingering echo from a changed dust distribution. The programme addresses two linked questions:

1. How long does warm-dust memory persist? Delayed radiative transfer can produce excess warm emission following a sharp nuclear drop, or an apparent deficit after an outburst [8,9]. Coupling historical WISE data with current MIRI spectra tests whether this thermal lag outlasts the primary transition.

2. Does dust replenish after the nucleus fades? The simplest null model assumes passive heating and cooling across a static dust distribution. By measuring the 5–28 μm continuum and silicate optical depths across recovering and fading sources [6–9]—benchmarked against rising baselines—we test whether structural clearing or replenishment is required.

From infrared echoes to dust evolution. WISE infrared echoes link changing-look transitions to intrinsic changes in nuclear power [10–13]. Their broadband colours blend dust temperature, emitting area, extinction, and host light [14–16]; spectroscopy supplies the missing diagnostics of the responding material.

The sublimation radius scales roughly as $L^{1/2}$ [17–19], but the dust boundary adjusts over years [6]. Mrk 590’s smaller dust radius after fading points to replenishment [7], whereas Mrk 817 shows temperature changes consistent with heating at a fixed boundary [20]. Fading and recovering nuclei test whether dust rebuilds when illumination drops; rising targets test its response to increased illumination. Does the infrared spectrum depend on the path taken to the present state?

Why MRS spectroscopy, rather than imaging? **Imaging measures how bright the dust is; our test also needs the shape of its spectrum.** Even spatially resolved MIRI colours mix the continuum, broad silicate features, PAH bands, and narrow lines. MRS measures the continuum beneath these features and the shapes of both the 9.7 and 18 μm silicate profiles, testing whether a warm excess is explained by delayed heating or requires a changed dust distribution [26,27]. Its integral-field data separate unresolved nuclear and resolved host spectra [28,29]. SPHEREx supplies the complementary 0.75–5 μm hot-dust spectrum [20–22]; MIRI low-resolution spectroscopy ends near 14 μm and misses the 18 μm feature. MRS provides the required 4.9–27.9 μm coverage [25] (Figure 2, top).

A sample selected to test dust memory. Our parent set combines catalogued changing-look

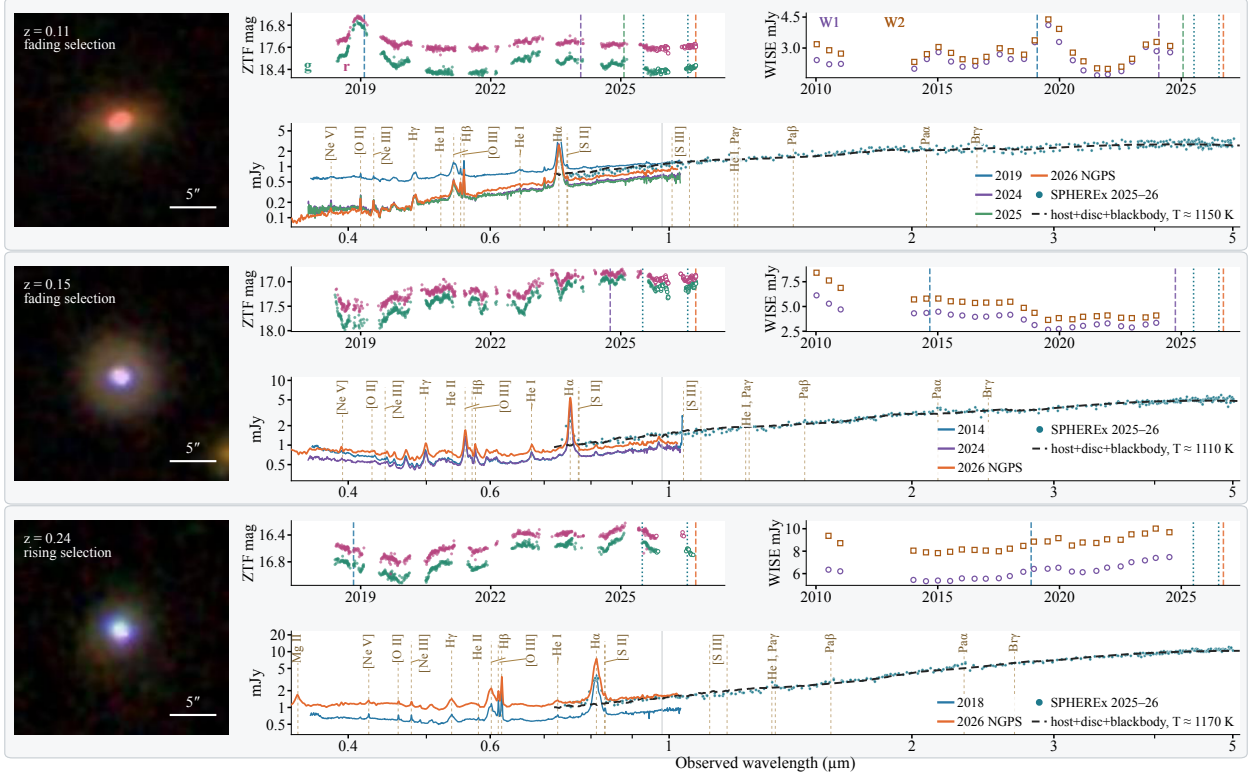


Figure 1: **Three examples of the rich histories assembled for the 24-target sample.** Each example combines ZTF and WISE light curves, archival and 2026 optical spectra and multiple SPHEREx visits, connecting nuclear activity to hot dust. **Top:** a 2018–2019 flare followed by fading; **middle:** an infrared decline with modest optical recovery; **bottom:** rising light curves and optical spectra. MIRI will test how much of this history the warm dust retains. SDSS images have 5'' bars. Open ZTF points are alerts through 2026 September; coloured ticks mark spectral epochs. Dashed curves show mean host+disc+blackbody fits to SPHEREx ($T_{\text{col}} \approx 1100\text{--}1200$ K, model-dependent); gold markers indicate emission lines. Wavelengths are observed-frame; spectra are in mJy.

AGN [30–33] and large-amplitude ZTF/WISE variables [5,34], across spectral types [31,35]; manifold learning groups related light-curve shapes [36]. We restrict selection to $z < 0.3$ to retain both silicate peaks, and prioritize recorded optical/infrared changes, archival or 2026 optical spectroscopy, and W3/W4 detections for exposure planning. The selected targets have ~ 10 years of W1/W2 monitoring, with 95th-to-5th-percentile flux ratios of 1.3–2.3 before host subtraction; all have archival W3/W4 fluxes of at least 1.9/6.4 mJy. SPHEREx adds the hot-dust spectrum. The final sample balances **12 histories dominated by fading and 12 by rising**. This comparison tests whether delayed heating explains both directions or whether fading nuclei require dust rebuilding. Flares and recoveries test whether nuclei at comparable current power retain different warm spectra because they followed different paths.

What we will measure and fit. We will model the extracted nuclear and surrounding spectra with the **SPIRIT spectral-decomposition framework** [15], illustrated in Figure 2 (top), fitting dust emission and extinction together with PAH profiles and measuring lines after continuum subtraction. The primary observables are **the rest-frame 8–13 μm dust luminosity, continuum slopes over 5.5–14.5 and 14.5–20 μm , and the 9.7/18 μm silicate strengths and profiles.** Slopes are measured from the fitted continuum; both intervals are covered for all 24 targets. The luminosity includes broad silicate emission or absorption, with PAHs and narrow lines removed. PAH fluxes and the 11.3/7.7 μm ratio constrain host contributions; [Ne v] and [Ne VI] provide supporting excitation diagnostics. We then compare the dust spectrum with response models driven by each recorded history: the continuum measures the warm excess or deficit, while the paired silicate profiles test whether the same dust distribution can explain it.

Why one visit works, even if the nucleus changes again. We observe the dust’s response to an already recorded event; success does not require catching a new transition. Warm dust responds to earlier illumination over a range of light-travel delays [8,23,24] (Figure 2, lower right). We evaluate each response model at the actual JWST date, using the full optical/infrared history and optical photometry nearest the visit to establish the current state. A further rise or decline enters that history and changes the predicted spectrum; it does not remove the source from the experiment. Gaps enter as uncertainties. Across 24 different event ages and directions, the snapshots test how long a warm imprint persists. The 2010 W3/W4 measurements add an earlier warm-dust anchor through a shared host model.

Measurement precision and interpretation. An illustrative shell model for the rising source in Figure 1 (bottom), using opacities from [37], gives conditional 1σ uncertainties of 5% on 8–13 μm flux and 0.08/0.04 on the silicate strengths, $S_\lambda = \ln(F_\nu/F_{\nu,\text{cont}})$, relative to a local power-law continuum. This assumes the requested S/N, correlated calibration errors of 3% in scale and tilt, and a smooth 10% host residual at rest 12 μm . Discrimination comes from the ensemble: one response prescription must reproduce 24 different histories, beyond what a single spectrum can establish.

Testing whether delayed heating is sufficient. Clumpy and disc-plus-wind distributions [38–40] define the null: unchanged dust heated by delayed illumination [8,9]. Dust replenishment or destruction [6,7,22], including a varying wind supply [41], motivates evolving distributions. Both families include light-travel delays and grain-size/composition uncertainty [42]. The warm residual is

$$\delta_{\text{warm}} = \log_{10} [L_{8-13\mu\text{m}}^{\text{MRS}} / L_{8-13\mu\text{m}}^{\text{fixed}}(t_{\text{MIRI}})],$$

relative to the *delayed fixed-distribution* prediction. An evolving distribution must improve the joint continuum/silicate fit beyond what fixed models, host templates, and calibration covariance allow [28,43]. Across event ages, these fits constrain replenishment timescales, the radial extent of responding dust, and how long mid-infrared diagnostics retain a past nuclear state [44,45].

■ Description of Observations (required for all GO and Survey)

Spectral coverage and sensitivity. Each target receives one MIRI/MRS science visit paired with a dedicated, noninterruptible background exposure using identical integration parameters [25,48].

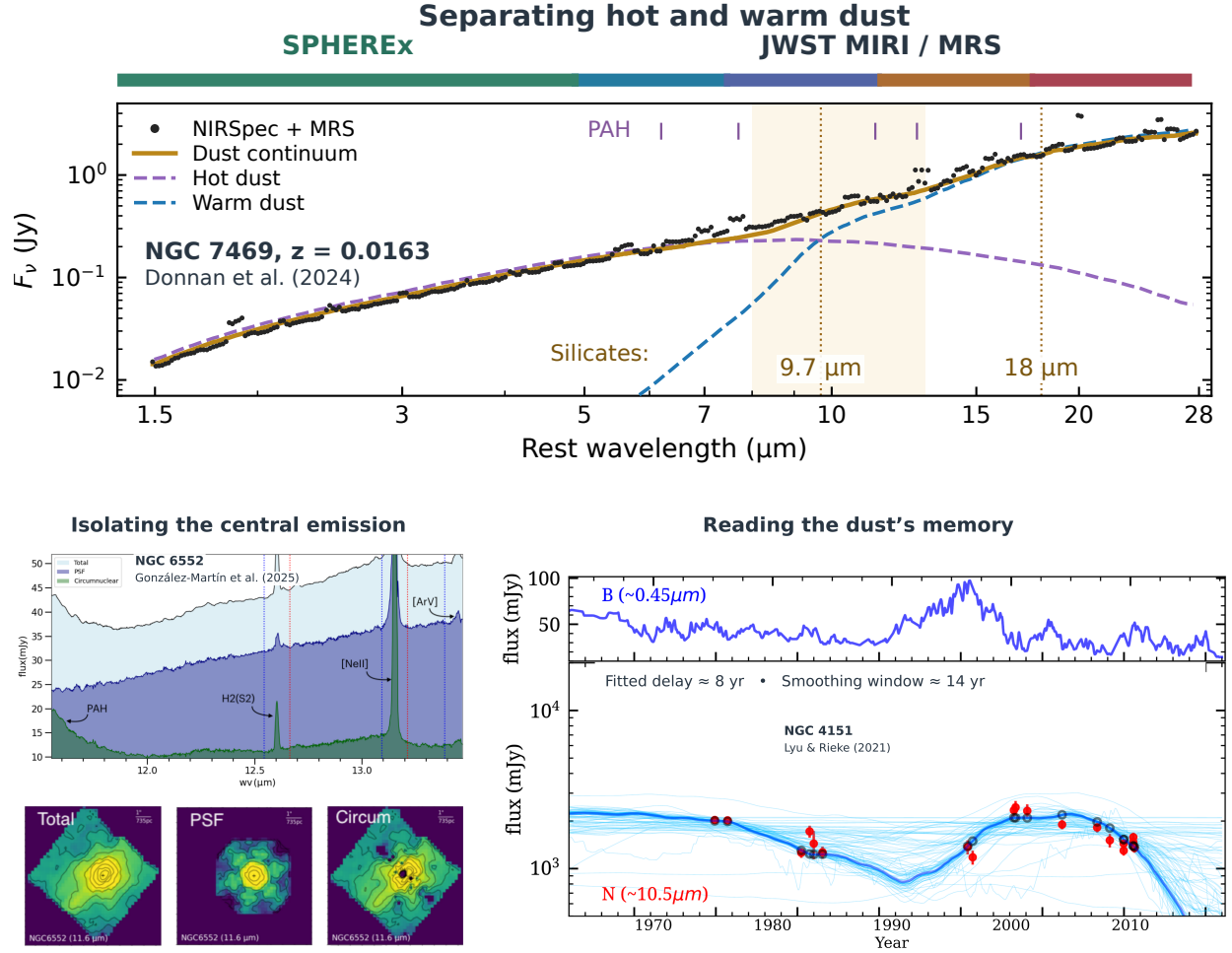


Figure 2: **The spectral, spatial, and temporal ingredients of the experiment, each already shown to work in a single AGN.** **Top—Spectral:** measuring the dust spectrum. The NIRSpec+MIRI/MRS spectrum of NGC 7469 (digitized from [15, Fig. 8]) separates cleanly into hot and warm dust components. The shaded 8–13 μm band provides the dust luminosity once PAH and line emission are subtracted, and the 9.7 and 18 μm silicate features add a spectral-shape diagnostic that imaging cannot supply. Bars mark MRS and, for comparison, SPHEREx coverage at $z = 0.0163$. **Lower left—Spatial:** isolating the nucleus. Spectra and 11.6 μm maps of NGC 6552 [28, Fig. 4] decompose the emission into nuclear (PSF) and circumnuclear components, the step that decides whether a nuclear warm excess survives host subtraction. **Lower right—Temporal:** reading the dust’s memory. In NGC 4151 the 10.5 μm emission (red points; light-blue curves are fitted responses) follows the optical light curve (blue) with an ~ 8 yr delay and additional smoothing [23, Fig. 15], preserving a record of earlier nuclear activity. Wavelengths are rest-frame in the top panel and observed-frame at lower left; the lower-right panel omits longer-wavelength data and ancillary fit labels. Our programme is the first to combine all three steps across a larger sample to test where delayed heating alone explains the mid-infrared response and where the dust distribution itself must evolve.

All three grating settings (SHORT, MEDIUM, LONG across Channels 1–4) are observed with four-point nuclear dithers. The 4.9–27.9 μm coverage includes both silicate peaks for all 24 targets; the primary rest-frame 8–13 μm diagnostic lies below observed 17 μm throughout the sample and does not require MIRI/MRS channel 4C.

Archival total-source W3/W4 fluxes span 1.9–48.4 and 6.4–82.6 mJy. We target S/N of 50 per $R = 100$ bin near rest 5 and 12 μm , and 30 at 18 μm (2% and 3.3% statistical errors, respectively). APT specifies the FULL/FASTR1 integrations, with extra setting-B exposure for the W3-faint J083826.50+371906.7. Equal-depth sky exposures preserve extended host emission [48]. Science-plus-sky integration totals 17.00 h; **the 67.0 h request** includes APT 2026.5.1 slew, acquisition, and exposure/setup overheads across 48 visits (24 science and 24 equal-depth sky), each with three grating settings and four-point dithers. This is a Medium programme [49].

For J083826.50+371906.7 at $z = 0.2111$, ETC/Pandea 2026.7 predicts S/N of 55/225/61 per $R = 100$ bin at rest 5/12/18 μm . These estimates assume a point source with $F_\nu \propto \lambda^2$, 0.95 mJy in W3 (half the archival total), the adopted sky spectrum, equal-depth sky exposures, and extraction radii of 0.3''/0.6''/0.9''. They establish statistical sensitivity under the adopted nuclear-flux model; host and calibration uncertainties enter the science inference separately. Acquisition uses F560W for the faintest sources and the neutral-density filter for brighter nuclei, with readouts yielding S/N above 50 without saturation over the adopted flux ranges [50].

Separating nuclear dust from the host. Each MRS cube measures the nucleus and surrounding host together. The PSF FWHM grows from approximately 0.3'' at 5 μm to 1'' at 25 μm , sampling roughly 0.1–4.5 kpc across the sample; channel fields span $3.2'' \times 3.7''$ to $6.6'' \times 7.7''$ [25]. Wavelength-dependent PSF fitting [51] separates unresolved and resolved emission [28] (Figure 2, lower left). Surrounding spectra constrain host templates with allowance for spatial variation. Joint modelling separates continuum, silicate, PAH, and H_2 contributions [15,47], allowing for AGN processing of PAHs [29]. Alternative extractions and host templates [16] bound the nuclear flux, including compact host light. [Ne VI] 7.7 and [Ne V] 14.3 μm are accessible for all targets [46]; the nominal range also includes [Ne V] 24.3 μm for 13 and [O IV] 25.9 μm for nine. These lines support the excitation analysis without assuming an instantaneous response to nuclear power.

We will integrate observed-frame MRS spectra through the WISE responses, using the measured spectral shape to calculate synthetic W3/W4 fluxes in the WISE calibration convention. Comparison with 2010 requires matching the WISE spatial response: constant host light cancels in a flux difference only for the same spatial contribution. WISE imaging and aperture photometry constrain emission beyond the MRS field; optical and 2MASS profiles guide stellar-host morphology. Models account for spectral coverage gaps and host light within the SPHEREx aperture. We propagate nuclear-flux bounds from alternative extractions and host templates, uncertainty in the aperture matching, W3/W4 calibration errors, and wavelength-dependent MRS calibration covariance [43].

Supplemental Information

Special requirements. Each science/sky pair is linked as a consecutive, noninterruptible sequence to obtain a contemporaneous background [48]. Sky fields are selected using WISE and 2MASS catalogues and images, with at least $25''$ clearance from catalogued sources. Twenty-three fields lie within 1–2 arcmin of their nuclei; the crowded field around J20145928+2523010 uses a $135''$ offset. Background targets are marked extended and use four-point extended-source dithers, with no target acquisition. No fixed-date constraints, coordinated parallels or joint-facility time are requested. APT’s Visit Planner returns scheduling windows for all 48 visits.

Duplications. A coordinate-based MAST search on 2026 September 26, including archived and planned JWST observations, found no overlapping observations within $120''$ of any of the 24 science positions. No potential MIRI/MRS duplications were identified.

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